Does In-Domain Masked Autoencoding Overtake ImageNet Initialisation Under Label Scarcity? A Pre-Registered Study Across Four Visual Domains

Namrata Pathak
M.Tech in Artificial Intelligence and Data Science
KIIT University, Bhubaneswar, India
namratapathak0410@gmail.com

***Abstract*—When labels are scarce, self-supervised pretraining on a target domain's own unlabelled images is often proposed as an alternative to ImageNet initialisation, especially for domains that look unlike ImageNet. We ask at which label budget masked-autoencoder (MAE) pretraining on the target images overtakes supervised ImageNet-1k initialisation of an identical Vision Transformer (ViT), and whether that crossover depends on a measured distance between the domain and ImageNet. In a pre-registered study we compare four initialisations (ResNet-18 from scratch, ViT-Tiny from scratch, ViT-Tiny from ImageNet, ViT-Tiny from in-domain MAE) on four domains (plant disease, satellite, objects, animals; 7,000 images each), with a leakage-safe split, five nested label budgets (5–100 %) and three seeds: 240 fine-tuning runs on a single 6 GB laptop GPU. ImageNet initialisation was best in all 20 dataset–budget cells and no crossover occurred. The gap narrowed steadily as labels increased in every domain, e.g. from -0.239 to -0.020 macro-F1 on PlantVillage and from -0.620 to -0.252 on Oxford Pets. It was smaller for the plant and satellite domains (mean -0.118 and -0.110) than for objects and animals (-0.317 and -0.489), but none of four measured domain distances predicted it significantly (exact $p \geq 0.33$). In-domain MAE improved on a ViT trained from scratch in all 20 cells (13 significant) and performed on par with a ResNet-18 trained from scratch. The ranking of methods by macro-F1 replicated across domains (Kendall's $W = 0.90$–1.00), whereas the ranking by calibration after temperature scaling did not ($W \leq 0.53$).**

***Index Terms*—self-supervised learning, masked autoencoders, vision transformers, transfer learning, label efficiency, domain distance, calibration, plant disease**

I. Introduction

Labelled data are expensive in specialised domains. Photographing a leaf costs nothing, but labelling it with the correct disease requires an expert; the same is true for land-cover maps and medical scans. Practitioners therefore start from pretrained weights. The default is supervised pretraining on ImageNet [1]: 1.28 million labelled photographs of everyday objects. The alternative made possible by self-supervised learning is to pretrain on the target domain's own unlabelled images, for example with a masked autoencoder (MAE) [2], which learns by reconstructing randomly hidden image patches and needs no labels.

The two options trade *amount* of prior knowledge against its *relevance*: ImageNet offers far more images, from a different visual world; in-domain MAE offers far fewer, from the right one. Several studies report that MAE pretraining on small target datasets helps Vision Transformers (ViTs) [3]–[6], and studies of contrastive learning found that in-domain pretraining can rival ImageNet [7]. What remains open is *when* one option overtakes the other as the number of labels grows, and whether this point can be predicted from how far the target domain is from ImageNet; Ericsson et al. [8] list the relation between domain distance and transfer performance as an open question.

We study this question under a controlled and pre-registered protocol. Our contributions are:

1) A controlled comparison of four initialisations on four visual domains, in which the three ViT arms share one architecture and differ only in their initial weights, with identical data, leakage-safe splits, nested label subsets and training protocol.
2) A crossover analysis of in-domain MAE versus ImageNet initialisation over five label budgets: no crossover occurs up to 100 % of ∼5,000 training labels, but the gap narrows steadily as labels increase in every domain.
3) Four measured distances between each domain and ImageNet (FID, OTDD, ImageNet confidence, linear-probe accuracy), computed before training. They disagree with one another, and none predicts the gap significantly.
4) A replicability analysis showing that method rankings by accuracy replicate across domains while rankings by calibration after temperature scaling do not.

The hypotheses and analysis plan were committed to a public repository before any training run. All 240 runs fit within 14.6 h on a laptop GPU, so the full study can be reproduced on modest hardware.

II. Related Work

Transfer from ImageNet. Better ImageNet models tend to transfer better [9], but the benefit shrinks for domains such as medical imaging, where ImageNet features are partly unnecessary [10]. ViTs [11] lack the locality bias of convolutional

networks [12] and depend strongly on pretraining. DeiT [13] showed that they can be trained on ImageNet-1k alone.

Self-supervised and in-domain pretraining. Ericsson et al. [8] benchmarked the transfer of self-supervised models and found that it varies with the target domain. Cole et al. [7] examined when contrastive learning works, varying the amount and domain of the pretraining data, for ResNet-50. MAE [2] masks 75 % of the patches and reconstructs their pixels. El-Nouby et al. [3] and Zhou et al. [4] showed that masked pretraining on the target data alone can be competitive, and Mao et al. [5] and Das et al. [6] used MAE to reduce the data hunger of ViTs. Hammoud et al. [14] studied how the diversity of the pretraining data matters. In plant pathology, deep networks reach high accuracy on PlantVillage [15], [16], and transformer-based self-supervised pretraining on plant images has been proposed [17]. None of these works locates the label budget at which in-domain MAE overtakes ImageNet across several domains.

Dataset distance and calibration. FID [18] compares feature distributions and OTDD [19] also takes the labels into account. Calibration of modern networks and temperature scaling were studied by Guo et al. [20] and revisited by Minderer et al. [21]. Sambyal et al. [22] examined calibration under limited data for medical images. Dehghani et al. [23] warn that conclusions drawn from a single benchmark may not replicate, which motivates our multi-domain design and rank statistics [24].

III. Method

A. Study design and arms

We compare four initialisations (*arms*), each fine-tuned on the same labelled subsets (Fig. 1). **A**: ResNet-18 [12] from random weights (11.2 M parameters). **C**: ViT-Tiny (DeiT-Tiny architecture [13], 12 blocks, width 192, 3 heads; 5.5 M parameters) initialised from supervised ImageNet-1k weights. **D**: the same ViT from random weights. **E**: the same ViT initialised from an MAE encoder pretrained on the target domain's own training images without labels. Because C, D and E share the architecture, their differences are attributable to the initial weights alone. E−C is the primary contrast; E−D isolates the effect of MAE, and E−A compares MAE with the built-in inductive bias of a convolutional network. ViT-Tiny, ResNet-18 and 160 px inputs were chosen so that the whole study fits a 48 h budget on one RTX 4050 Laptop GPU (6 GB).

B. Data, split and label budgets

We use one dataset per domain (Table I): PlantVillage [15] (plant disease, colour), EuroSAT RGB [25] (satellite land cover), CIFAR-10 [26] (objects) and Oxford-IIIT Pets [27] (animal breeds). From each we draw a class-stratified sample of 7,000 images, so that every domain has the same amount of unlabelled pretraining data and the same label counts. Every image is resized (shorter side to 160 px, bicubic) and centre-cropped to 160×160; CIFAR-10 and EuroSAT are therefore upsampled.

Near-duplicate images (e.g. several photographs of the same leaf) would inflate test scores if they were spread across splits. We compute a 256-bit difference hash per image, join images with Hamming distance ≤ 10 into groups (union-find), and split 70/15/15 with `StratifiedGroupKFold` (seed 42), so that no group crosses a split. Label budgets are 5, 10, 20, 50 and 100 % of each class's training images (at least one per class), i.e. about 250, 500, 1,000, 2,500 and 5,000 labelled images. Subsets are nested (the 5 % subset is contained in the 10 % subset, and so on) and identical for all arms within a seed. Each seed (0, 1, 2) fixes the labelled subset, the initialisation of new layers, the data order and E's MAE pretraining.

C. Domain distance

Each image is embedded with the frozen ImageNet-1k DeiT-Tiny used by arm C (class-token features, 192-d, 160 px). The reference is ImageNetV2 matched-frequency [28] (10,000 images, 1,000 classes). We compute four distances on the training split: (i) FID [18] between the two feature distributions; (ii) OTDD [19], an optimal-transport distance whose ground cost includes a label-to-label term (diagonal-Gaussian class model, 64-d PCA, 1,500 samples per side, solved with POT [29]); (iii) ImageNet confidence, the mean maximum softmax probability of the 1,000-way ImageNet classifier on the domain's images (0.458 on ImageNetV2 itself); and (iv) the validation accuracy of a logistic-regression linear probe on the frozen features. Higher FID or OTDD and lower confidence or probe accuracy indicate a domain farther from ImageNet.

D. MAE pretraining

For arm E we pretrain an MAE [2] whose encoder is the ViT-Tiny of arms C and D. Encoder position embeddings are initialised to 2-D sine–cosine values. 75 % of the 100 patches are masked; a 2-block, 128-wide decoder (4 heads) reconstructs the normalised pixels of the masked patches (norm-pixel loss). We train for 800 epochs over the ∼5,000 unlabelled training images (15,200 steps, batch 256, AdamW with $\beta = (0.9, 0.95)$, base learning rate 1.5×10^{-4}, weight decay 0.05, 40 warm-up epochs, cosine decay, random resized crop 0.2–1 and horizontal flip). Pretraining is repeated for each seed (12 pretrainings in total), takes ∼21.5 min each, and uses neither labels nor validation or test images.

E. Fine-tuning protocol

All arms follow the protocol in Table II, in which the number of training steps grows with the size n of the labelled set. Validation macro-F1 is evaluated 25 times per run, training stops after 8 evaluations without improvement, and the best-validation weights are restored before the single test evaluation. For the pretrained ViTs (C, E) we use layer-wise learning-rate decay of 0.75 [30], so earlier blocks change less. The learning rate is chosen per dataset and arm from three values, on validation macro-F1, at the 20 % budget with seed 0, and then reused for every budget and seed.

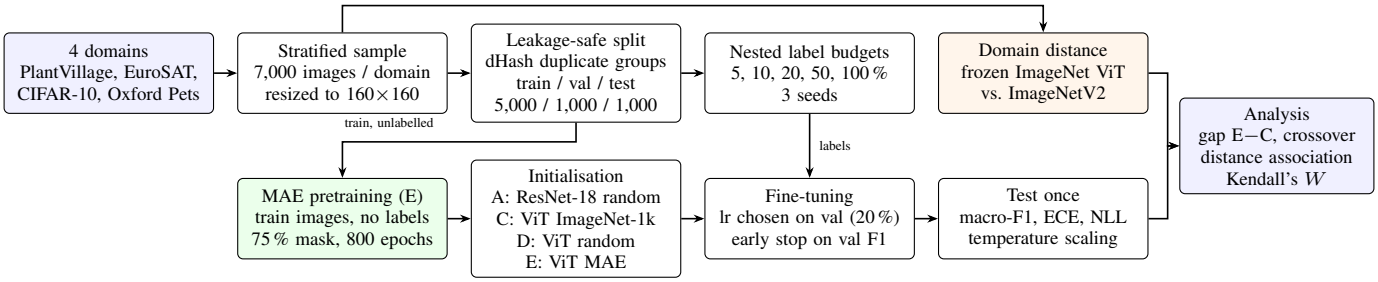


Fig. 1. Study pipeline. The same pipeline is run independently for each of the four domains. Only arm E's MAE pretraining sees the (unlabelled) training images before fine-tuning; validation and test images are never used for pretraining, and test images are evaluated once per run.

TABLE I
DATASETS, LEAKAGE-SAFE SPLITS AND DOMAIN DISTANCES TO IMAGENET (COMPUTED BEFORE TRAINING). ARROWS GIVE THE DIRECTION THAT MEANS "FARTHER FROM IMAGENET".

Dataset	Domain	Classes	Native px	Train/Val/Test	Dup. groups	FID ↑	OTDD ↑	ImageNet conf. ↓	Linear probe ↓
PlantVillage	plant	38	256	4999/1000/1001	0	433	30.6	0.30	0.90
EuroSAT	satellite	10	64	5000/1000/1000	1	318	26.4	0.14	0.86
CIFAR-10	objects	10	32	5000/1000/1000	0	267	28.3	0.29	0.83
Oxford Pets	animals	37	var.	5000/1000/1000	34	273	29.3	0.51	0.86

TABLE II
FINE-TUNING PROTOCOL (IDENTICAL FOR ALL ARMS UNLESS NOTED).

Optimiser	AdamW [31], weight decay 0.05
Batch / precision	128 / bf16, gradient clipping 1.0
Schedule	5 % linear warm-up, cosine decay
Steps	$\mathrm{clip}(100\lceil n/128\rceil, 600, 4000)$
Early stopping	25 val. checks, patience 8, restore best
Augmentation	random resized crop 0.35–1, flip, brightness/contrast ±20 %
ViT details	drop-path 0.1, average-pooled tokens
Layer-wise lr decay	0.75 (C, E only)
LR grid, scratch (A, D)	$\{5\times10^{-4}, 10^{-3}, 2\times10^{-3}\}$
LR grid, pretrained (C, E)	$\{2.5\times10^{-4}, 5\times10^{-4}, 10^{-3}\}$

F. Metrics and statistical analysis

The primary metric is test macro-F1. We also report the expected calibration error (ECE, 15 equal-width bins [20], [32]) before and after temperature scaling, with the temperature fitted on the validation set. For each dataset and budget we compute the paired gap E−C over seeds with a 95 % t-interval. The crossover budget is the sign change of the mean gap, interpolated on $\log_{10}$(budget); without a sign change it is reported as censored (>100 % or <5 %), and a crossover counts as supported only if the intervals exclude zero on both sides. Each dataset is summarised by the mean gap over the log-budget axis (trapezoidal rule). Its association with each distance uses Spearman's ρ with an exact permutation p-value and a leave-one-dataset-out linear prediction compared with predicting the mean. With four datasets the smallest attainable two-sided p is $2/4! = 0.083$, so this analysis was declared exploratory in advance. Replicability is measured by Kendall's W [33] and the Friedman test of the arm ranks across datasets, per budget.

G. Pre-registration

Hypotheses H1–H5 (Table VII), the protocol and the analysis plan were committed to the repository before any training run. One amendment, also made before training and recorded with its reason, raised MAE pretraining from 400 to 800 epochs after a speed benchmark showed that the time budget allowed the standard MAE schedule. The domain distances were recorded in the pre-registration before training as well.

IV. RESULTS

All 240 fine-tuning runs completed (9.3 h of fine-tuning, 1.0 h of learning-rate selection and ∼4.3 h of MAE pretraining; 14.6 h wall-clock, peak GPU memory 3.4 GB).

A. ImageNet initialisation wins everywhere; no crossover

Fig. 2 and Table III show that arm C achieved the highest mean macro-F1 in all 20 dataset–budget cells. The paired gap E−C (Fig. 3) was negative everywhere, with 95 % intervals excluding zero in all 20 cells. Hence the crossover budget is censored at >100 % for every dataset (Table IV), and H1 is supported only in its weaker form: the gap *closes* with more labels in every domain but never reverses. The narrowing was steady: on PlantVillage the gap shrank from -0.239 at 5 % to -0.020 at 100 % (95 % CI $[-0.037, -0.003]$), and on EuroSAT from -0.183 to -0.049. On CIFAR-10 ($-0.443 \rightarrow -0.196$) and Oxford Pets ($-0.620 \rightarrow -0.252$) the gap remained large.

B. Does domain distance predict the gap?

The mean gap was much smaller for PlantVillage (-0.118) and EuroSAT (-0.110) than for CIFAR-10 (-0.317) and Oxford Pets (-0.489). The latter two contain object and animal categories that ImageNet covers extensively; Pets is the only domain on which the ImageNet classifier is more confident

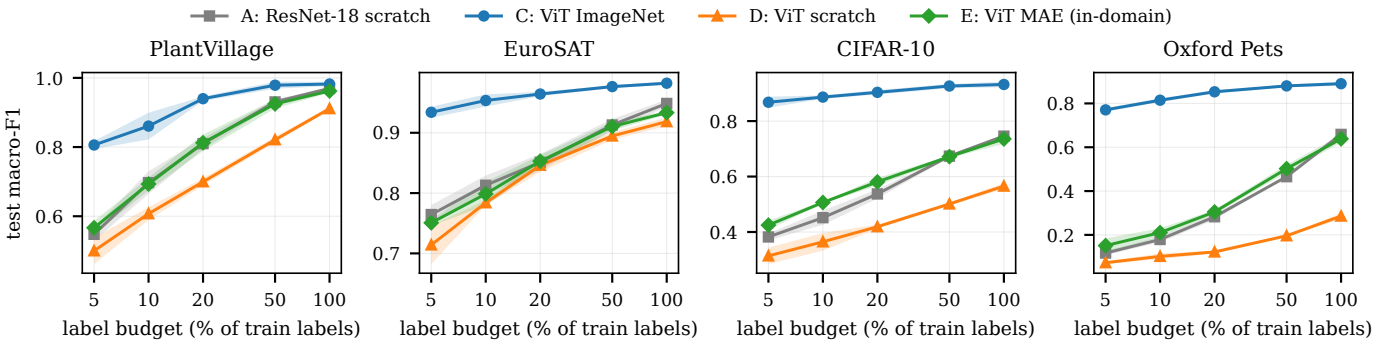


Fig. 2. Test macro-F1 versus label budget (mean of 3 seeds; shading ±1 s.d.). The ImageNet-initialised ViT (C) is best everywhere; in-domain MAE (E) lies above the scratch ViT (D) and close to the scratch ResNet-18 (A).

TABLE III
TEST MACRO-F1 (MEAN ± S.D. OVER 3 SEEDS). BOLD: BEST ARM. LAST COLUMN: PAIRED GAP E−C; *95 % CI EXCLUDES 0.

	%	A: CNN scr.	C: ViT IN	D: ViT scr.	E: ViT MAE	E−C
PlantVillage	5	0.548 ±0.019	**0.806 ±0.012**	0.500 ±0.038	0.567 ±0.020	−0.239*
	10	0.697 ±0.033	**0.861 ±0.039**	0.607 ±0.019	0.693 ±0.019	−0.167*
	20	0.810 ±0.016	**0.940 ±0.006**	0.700 ±0.010	0.812 ±0.024	−0.127*
	50	0.930 ±0.007	**0.979 ±0.010**	0.821 ±0.008	0.925 ±0.014	−0.054*
	100	0.970 ±0.003	**0.982 ±0.006**	0.911 ±0.005	0.962 ±0.005	−0.020*
EuroSAT	5	0.764 ±0.016	**0.934 ±0.009**	0.714 ±0.032	0.751 ±0.005	−0.183*
	10	0.813 ±0.016	**0.954 ±0.010**	0.784 ±0.004	0.799 ±0.020	−0.155*
	20	0.851 ±0.014	**0.964 ±0.004**	0.846 ±0.011	0.853 ±0.010	−0.111*
	50	0.913 ±0.010	**0.977 ±0.003**	0.895 ±0.006	0.910 ±0.007	−0.066*
	100	0.949 ±0.006	**0.982 ±0.003**	0.918 ±0.007	0.933 ±0.003	−0.049*
CIFAR-10	5	0.382 ±0.012	**0.868 ±0.020**	0.314 ±0.030	0.425 ±0.017	−0.443*
	10	0.452 ±0.026	**0.887 ±0.006**	0.365 ±0.032	0.506 ±0.004	−0.380*
	20	0.537 ±0.017	**0.904 ±0.008**	0.419 ±0.007	0.581 ±0.014	−0.323*
	50	0.674 ±0.004	**0.926 ±0.008**	0.501 ±0.005	0.672 ±0.005	−0.255*
	100	0.746 ±0.006	**0.932 ±0.010**	0.565 ±0.011	0.736 ±0.018	−0.196*
Oxford Pets	5	0.117 ±0.010	**0.771 ±0.008**	0.073 ±0.006	0.151 ±0.033	−0.620*
	10	0.179 ±0.013	**0.815 ±0.008**	0.102 ±0.013	0.210 ±0.020	−0.604*
	20	0.283 ±0.014	**0.853 ±0.007**	0.122 ±0.007	0.305 ±0.009	−0.548*
	50	0.466 ±0.001	**0.880 ±0.006**	0.196 ±0.004	0.502 ±0.018	−0.378*
	100	0.659 ±0.015	**0.890 ±0.009**	0.286 ±0.014	0.638 ±0.017	−0.252*

TABLE IV
PER-DATASET SUMMARY OF THE PAIRWISE GAPS (MACRO-F1). MEAN GAP: AVERAGE OF E−C OVER THE LOG-BUDGET AXIS. RANGES ARE OVER THE FIVE BUDGETS; "SIG." COUNTS BUDGETS WHOSE 95 % CI EXCLUDES 0.

	E−C		E−D		E−A	
Dataset	mean gap	crossover	range	sig.	range	sig.
PlantVillage	−0.118	> 100 %	+0.051 to +0.113	4/5	−0.008 to +0.019	0/5
EuroSAT	−0.110	> 100 %	+0.007 to +0.037	0/5	−0.015 to +0.002	0/5
CIFAR-10	−0.317	> 100 %	+0.111 to +0.171	5/5	−0.010 to +0.055	1/5
Oxford Pets	−0.489	> 100 %	+0.078 to +0.353	4/5	−0.020 to +0.037	1/5

(0.51) than on ImageNetV2 itself (0.46). However, the four distance measures do not agree on which domains are far from ImageNet (Table I): FID ranks PlantVillage farthest, ImageNet confidence ranks EuroSAT farthest, and OTDD ranks EuroSAT *closest*. Accordingly, their association with the mean gap is inconsistent (Fig. 4, Table V). FID ($\rho = +0.6$) and ImageNet confidence ($\rho = -0.8$) point in the hypothesised direction, OTDD points in the opposite direction ($\rho = -0.4$) and the

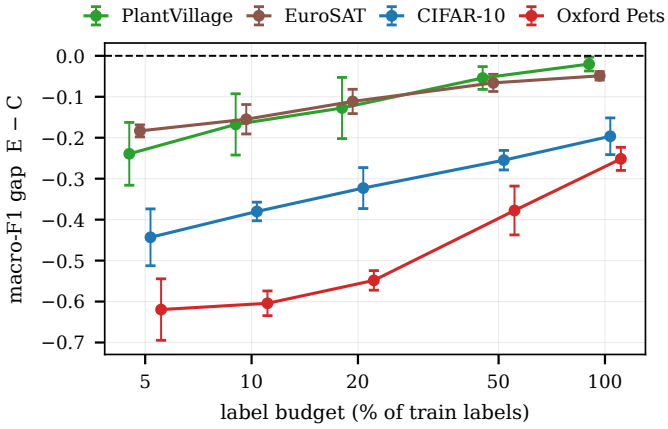


Fig. 3. Paired macro-F1 gap between in-domain MAE (E) and ImageNet (C) initialisation, with 95 % t-intervals over 3 seeds. Below the dashed line ImageNet is ahead. The gap narrows with the label budget in every domain but does not reach zero.

TABLE V
ASSOCIATION BETWEEN DOMAIN DISTANCE AND MEAN GAP E−C (EXPLORATORY, $n = 4$). "PRED.": SIGN OF ρ EXPECTED UNDER H2. LOO: MEAN ABSOLUTE ERROR OF A LEAVE-ONE-DATASET-OUT LINEAR PREDICTION VS. PREDICTING THE MEAN OF THE OTHER DATASETS.

Distance	Pred.	ρ	exact p	LOO error	baseline
FID	+	+0.600	0.417	0.288	0.193
OTDD	+	−0.400	0.750	0.350	0.193
ImageNet confidence	−	−0.800	0.333	0.124	0.193
Linear-probe accuracy	−	+0.000	1.000	0.205	0.193

linear probe shows no association. No p-value falls below 0.33, and only ImageNet confidence predicts the held-out dataset better than the mean (absolute error 0.124 vs. 0.193). H2 is therefore not supported at the level of evidence that four domains can provide.

C. MAE versus training from scratch

In-domain MAE improved on the identically trained scratch ViT in all 20 cells, significantly in 13 (Table IV), so H3 is mostly supported. The benefit was largest on Oxford Pets, where it grew with the budget to +0.353 at 100 % (0.638 vs. 0.286), and on CIFAR-10 (+0.111 to +0.171). On EuroSAT

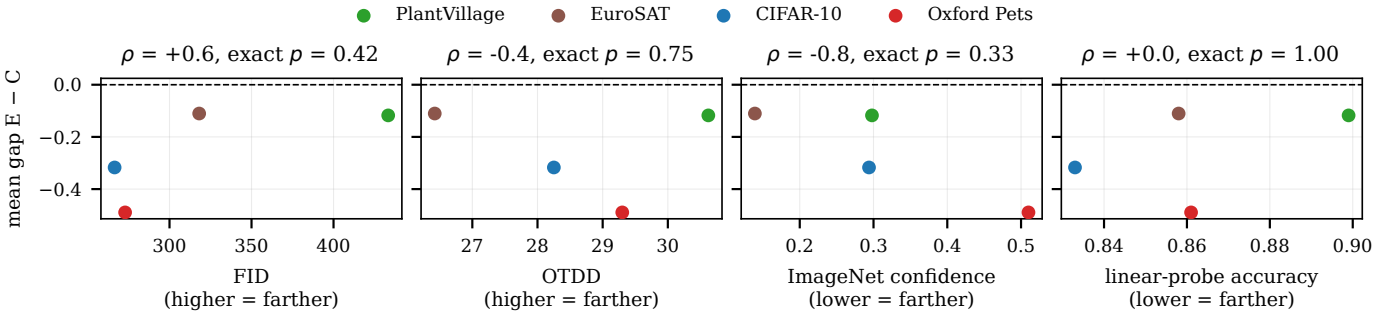


Fig. 4. Mean gap E−C (averaged over log-budget) against the four domain distances measured before training. Exact two-sided permutation p-values; with four datasets the minimum attainable p is 0.083.

TABLE VI
AGREEMENT OF ARM RANKINGS ACROSS THE FOUR DATASETS: KENDALL'S W (1 = IDENTICAL RANKINGS) AND FRIEDMAN p, PER BUDGET.

	macro-F1		ECE		ECE after TS	
Budget %	W	p	W	p	W	p
5	0.925	0.011	0.775	0.026	0.050	0.896
10	0.900	0.013	0.900	0.013	0.075	0.825
20	1.000	0.007	0.375	0.212	0.525	0.098
50	0.925	0.011	0.700	0.038	0.450	0.145
100	1.000	0.007	0.550	0.086	0.225	0.440

it was small (+0.007 to +0.037) and not significant at any budget; there the scratch ViT was already strong (0.918 at 100 %). Against the scratch ResNet-18 (E−A), all differences stayed within −0.020 and +0.055, and only 2 of 20 were significant (CIFAR-10 at 5 %, Pets at 20 %). At this data scale, MAE pretraining thus gives a ViT roughly what the convolutional inductive bias gives a ResNet, but not more.

D. Replicability of rankings and calibration

The ranking of the four arms by macro-F1 was almost identical across domains: Kendall's W was between 0.90 and 1.00 at every budget (Friedman $p \leq 0.013$; Table VI), with the order C > E ≈ A > D. H4, which predicted that the best arm would differ between domains, is therefore rejected.

Calibration behaved differently (Fig. 5). Before temperature scaling, C also had the lowest ECE in 17 of 20 cells (mean ECE 0.052 vs. 0.119 for E, 0.154 for D and 0.171 for A). Temperature scaling reduced the ECE of all arms to 0.024–0.032 on average, and the ranking of arms by post-scaling ECE no longer agreed across datasets ($W = 0.05$–0.53, all Friedman $p \geq 0.098$); the scratch ViT was best calibrated in 7 cells. H5 is thus partly supported: the ranking by calibration departs from the ranking by accuracy once a simple post-hoc correction is applied.

V. DISCUSSION

Why does ImageNet win? ImageNet-1k supplies about 250 times more images than our in-domain pool, together with labels that force the network to learn shapes, parts and textures. Such generic features transfer even to leaves and satellite

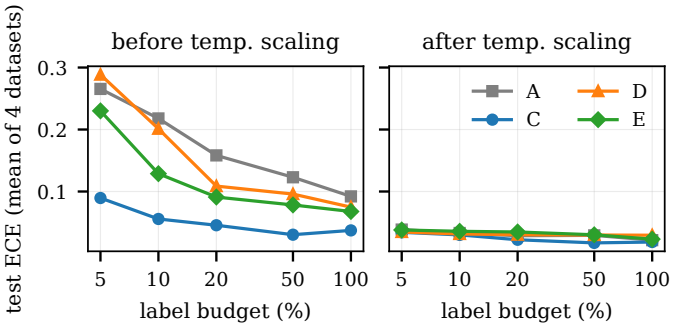


Fig. 5. Test ECE versus label budget, averaged over the four datasets, before and after temperature scaling (temperature fitted on validation data).

TABLE VII
PRE-REGISTERED HYPOTHESES AND OUTCOMES.

	Hypothesis	Outcome
H1	C ahead at small budgets; E closes or reverses the gap at larger budgets	**Partly**: gap closes in all 4 domains; no crossover (>100 %)
H2	Gap E−C smaller for domains farther from ImageNet	**Not supported**: direction matches for 2 of 4 distances, none significant
H3	E > D at every budget in every domain	**Mostly**: 20/20 positive, 13/20 significant; none on EuroSAT
H4	Best arm differs across domains and budgets ($W < 1$)	**Rejected**: C best in 20/20 cells, $W = 0.90$–1.00
H5	Ranking by ECE differs from ranking by macro-F1	**Partly**: similar before, different after temperature scaling

tiles. The ImageNet-initialised ViT also converged fastest: it triggered early stopping in 67 % of its runs, compared with 20 % for D and E and 8 % for A. MAE, by contrast, learns from ∼5,000 images by reconstructing pixels. Its reconstructions (Fig. 6) recover the global layout, for instance the outline and colour of a leaf, but smooth away small, high-frequency details, among them the lesions that distinguish many PlantVillage classes. This is consistent with the gap shrinking as the labelled set grows: fine-tuning must then supply the fine-grained detail that the reconstruction objective did not emphasise.

Where does in-domain MAE come close? The gap was smallest in the two domains that are visually least like ImageNet's object-centric photographs (PlantVillage, EuroSAT) and largest where the content overlaps with ImageNet cate-

gories (CIFAR-10, Pets). This matches the intuition behind H2. However, the measured distances do not capture it consistently, and a single scalar distance appears too coarse to predict the crossover. EuroSAT illustrates a further point: it had the lowest final MAE reconstruction loss (0.19, vs. 0.27–0.54 elsewhere) yet the smallest benefit of MAE over a scratch ViT, so a low reconstruction loss does not by itself indicate a useful representation.

Practical implication. With a few thousand unlabelled images and a small ViT, ImageNet initialisation remains the better default at every label budget. In domains far from ImageNet, the advantage at full labels can be small (two macro-F1 points on PlantVillage), and in-domain MAE remains a clear improvement over training a ViT from scratch. Because accuracy rankings replicated across domains but calibration rankings did not, calibration should be evaluated per domain rather than inferred from accuracy.

VI. Limitations

The study was designed for one laptop GPU and 48 h, which bounds its scope. (i) We use ViT-Tiny at 160 px and 7,000 images per domain. Larger models, higher resolutions and larger unlabelled pools favour MAE in the literature [2], so a crossover may exist beyond our range. (ii) Four domains give at most an exact p of 0.083 for a rank correlation, so H2 could only be tested in an exploratory way. (iii) Three seeds give wide intervals for small effects. (iv) The learning rate was tuned once per dataset and arm, at the 20 % budget, and for 11 of 16 dataset–arm pairs the chosen value lies at an edge of its three-value grid. A wider grid could shift some results, including the best rate for E on PlantVillage and Pets. (v) MAE used one fixed recipe (no search over masking ratio or schedule), and the DeiT weights for C were trained at 224 px, so their position embeddings were interpolated to 160 px. (vi) CIFAR-10 and EuroSAT were upsampled from 32 and 64 px. (vii) ImageNetV2 stands in for ImageNet in the distance measures, and C additionally benefits from far more pretraining compute than E; we do not match compute between the arms.

VII. Conclusion

We asked when in-domain MAE overtakes ImageNet initialisation for a small ViT as labels become more plentiful. Across four domains, under a pre-registered and leakage-safe protocol, the answer at a scale of 5,000 images is: not within 5–100 % of the labels. ImageNet initialisation was best in all 20 settings. Its advantage shrank steadily as labels increased and was smallest for plant-disease and satellite images (two macro-F1 points on PlantVillage at full labels), but none of four standard domain distances predicted this reliably. In-domain MAE consistently improved on a ViT trained from scratch and matched a CNN trained from scratch. Accuracy rankings replicated across domains; calibration rankings did not. Natural next steps are larger unlabelled pools, larger ViTs and more domains, which would allow the relation between domain distance and the crossover budget to be tested with adequate statistical power.

Reproducibility

Code, configuration, pre-registration, split files and all run-level results are available at
https://github.com/namratapathak0410-pixel/Label-Efficient-Representation-Learning-SSL.
Models were built with PyTorch and timm [34]. All figures and tables of this paper are generated from the result files by `paper/make_figures.py`.

References

[1] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2009, pp. 248–255.

[2] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 16 000–16 009.

[3] A. El-Nouby, G. Izacard, H. Touvron, I. Laptev, H. Jégou, and E. Grave, "Are large-scale datasets necessary for self-supervised pre-training?" *arXiv preprint arXiv:2112.10740*, 2021.

[4] L. Zhou, H. Liu, J. Bae, J. He, D. Samaras, and P. Prasanna, "Self pre-training with masked autoencoders for medical image classification and segmentation," *arXiv preprint arXiv:2203.05573*, 2022.

[5] J. Mao, H. Zhou, X. Yin, Y. Chang, B. Nie, and R. Xu, "Masked autoencoders are effective solution to transformer data-hungry," *arXiv preprint arXiv:2212.05677*, 2022.

[6] S. Das, T. Jain, D. Reilly, P. Balaji, S. Karmakar, S. Marjit, X. Li, A. Das, and M. S. Ryoo, "Limited data, unlimited potential: A study on ViTs augmented by masked autoencoders," in *Proc. IEEE/CVF Winter Conf. Applications of Computer Vision (WACV)*, 2024.

[7] E. Cole, X. Yang, K. Wilber, O. Mac Aodha, and S. Belongie, "When does contrastive visual representation learning work?" in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 14 755–14 764.

[8] L. Ericsson, H. Gouk, and T. M. Hospedales, "How well do self-supervised models transfer?" in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 5414–5423.

[9] S. Kornblith, J. Shlens, and Q. V. Le, "Do better ImageNet models transfer better?" in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 2661–2671.

[10] M. Raghu, C. Zhang, J. Kleinberg, and S. Bengio, "Transfusion: Understanding transfer learning for medical imaging," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[11] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.

[12] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[13] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *Proc. Int. Conf. Machine Learning (ICML)*, 2021, pp. 10 347–10 357.

[14] H. A. A. K. Hammoud, T. Das, F. Pizzati, P. Torr, A. Bibi, and B. Ghanem, "On pretraining data diversity for self-supervised learning," in *Proc. European Conf. Computer Vision (ECCV)*, 2024.

[15] D. P. Hughes and M. Salathé, "An open access repository of images on plant health to enable the development of mobile disease diagnostics," *arXiv preprint arXiv:1511.08060*, 2015.

[16] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers in Plant Science*, vol. 7, p. 1419, 2016.

[17] Z. Wang, R. Wang, M. Wang, T. Lai, and M. Zhang, "Self-supervised transformer-based pre-training method with general plant infection dataset," *arXiv preprint arXiv:2407.14911*, 2024.

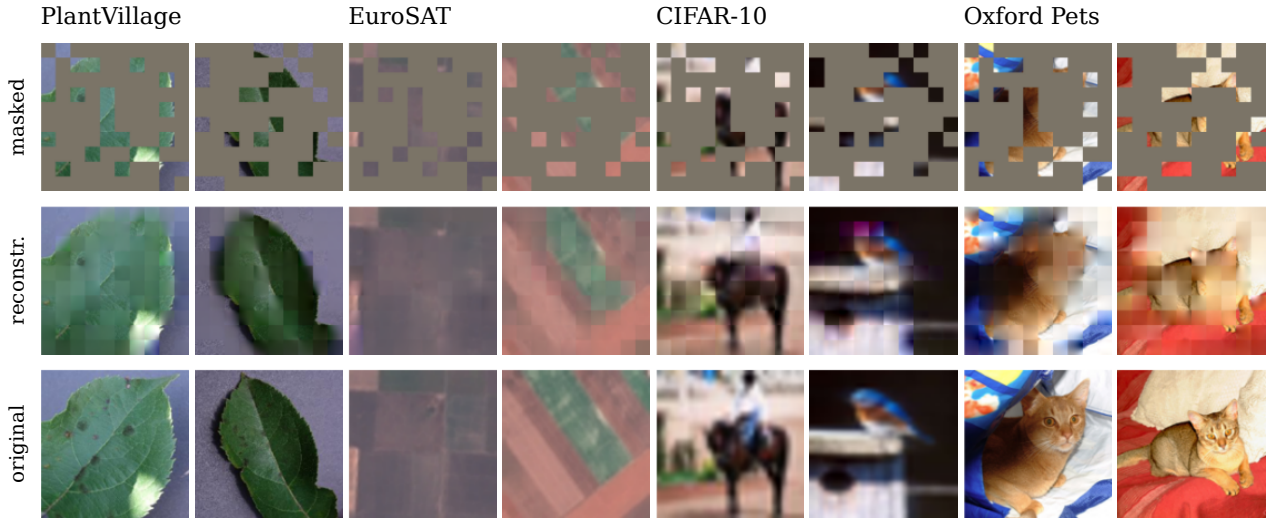


Fig. 6. MAE reconstructions (seed 0, two training images per domain). Top: input with 75 % of the patches masked; middle: reconstruction; bottom: original. The global layout is recovered, but fine detail such as leaf lesions is smoothed away.

[18] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[19] D. Alvarez-Melis and N. Fusi, "Geometric dataset distances via optimal transport," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[20] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. Int. Conf. Machine Learning (ICML)*, 2017, pp. 1321–1330.

[21] M. Minderer, J. Djolonga, R. Romijnders, F. Hubis, X. Zhai, N. Houlsby, D. Tran, and M. Lucic, "Revisiting the calibration of modern neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[22] A. S. Sambyal, U. Niyaz, N. C. Krishnan, and D. R. Bathula, "Understanding calibration of deep neural networks for medical image classification," *arXiv preprint arXiv:2309.13132*, 2023.

[23] M. Dehghani, Y. Tay, A. A. Gritsenko, Z. Zhao, N. Houlsby, F. Diaz, D. Metzler, and O. Vinyals, "The benchmark lottery," *arXiv preprint arXiv:2107.07002*, 2021.

[24] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *J. Machine Learning Research*, vol. 7, pp. 1–30, 2006.

[25] P. Helber, B. Bischke, A. Dengel, and D. Borth, "EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification," *IEEE J. Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 12, no. 7, pp. 2217–2226, 2019.

[26] A. Krizhevsky, "Learning multiple layers of features from tiny images," University of Toronto, Tech. Rep., 2009.

[27] O. M. Parkhi, A. Vedaldi, A. Zisserman, and C. V. Jawahar, "Cats and dogs," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2012, pp. 3498–3505.

[28] B. Recht, R. Roelofs, L. Schmidt, and V. Shankar, "Do ImageNet classifiers generalize to ImageNet?" in *Proc. Int. Conf. Machine Learning (ICML)*, 2019, pp. 5389–5400.

[29] R. Flamary *et al.*, "POT: Python optimal transport," *J. Machine Learning Research*, vol. 22, no. 78, pp. 1–8, 2021.

[30] H. Bao, L. Dong, S. Piao, and F. Wei, "BEiT: BERT pre-training of image transformers," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022.

[31] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2019.

[32] M. P. Naeini, G. F. Cooper, and M. Hauskrecht, "Obtaining well calibrated probabilities using Bayesian binning," in *Proc. AAAI Conf. Artificial Intelligence*, 2015, pp. 2901–2907.

[33] M. G. Kendall and B. Babington Smith, "The problem of m rankings," *The Annals of Mathematical Statistics*, vol. 10, no. 3, pp. 275–287, 1939.

[34] R. Wightman, "PyTorch image models," https://github.com/huggingface/pytorch-image-models, 2019.